Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
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Project Title:	Computational Models of the Cardiovascular System and Its Response to Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
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No. of Post Docs:	3	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
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Key Personnel Changes/Previous PI:	Kamm, Roger		
COI Name (Institution):	Kamm, Roger (Massachusetts Institute of Technology)		
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Experimental studies of the cardiovascular system in humans, and even in laboratory animals, are necessarily limited in scope because of restrictions on the types of measurements that can be made. Often, the true parameter of interest cannot be measured directly so it must be inferred from other measures. Even when appropriate measurements can be made, the cause of a particular observation may not be evident because of the complexity of the interactions between the numerous components of the system. These and other issues can often be addressed more effectively with the aid of a computational model that simulates the critical components and behaviors of the cardiovascular system. Models depend upon experiments for refinement and specification of their parameters, but also illuminate and enhance the interpretation of experimental results. We view the experiments and computational models as highly synergistic in that the value of one is greatly enhanced by the existence of the other. It could be argued that these are not merely advantages, but essential aspects of a study of orthostatic intolerance.
The primary objectives of this project are to develop a general, modular model of cardiovascular function that contains

Task Description:	the essential features associated with the effects of gravity, and to use this model to examine the short term effects of changes in posture before and after exposure to the microgravity environment. The objective of the currently funded project was to extend a previously developed computational model of the cardiovascular system and to use this model to investigate the short-term (0--5 mins) beat-to-beat hemodynamic response of the cardiovascular system to abrupt orthostatic stress --- such as head-up tilt (HUT) or lower body negative pressure (LBNP). The project aimed at facilitating the understanding of the physiology and treatment (prevention) of orthostatic intolerance in post-flight astronauts. We proposed to: - Enhance the current version of the cardiovascular simulation to better represent the short-term effects of abrupt orthostatic stress. - Verify the model and use it to investigate and evaluate various hypotheses of orthostatic intolerance, and to predict the effects of countermeasures. This objective required extensive collection and archiving of experimental data from collaborators. - Complete, document, and disseminate to other investigators a form of the model implemented in JAVA. - Apply the cardiovascular model to the clinical problem of intelligent patient monitoring with particular emphasis on establishing an enhanced research database of multiparameter hemodynamic data from intensive care patients.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	While the primary purpose of the proposed modeling effort is to provide a tool for use in studies of orthostatic hypotension and countermeasures, the resulting models will have value in other settings as well. As a clinical tool, models enhance the quality of measurement, both in the context of remote clinical monitoring in the space environment and in the ground-based clinical setting. Continuous hemodynamic monitoring in the ICU could be augmented by the use of models capable of tracking a particular patient's status. The challenge of rapidly assessing the pathophysiologic state of an unstable patient is made more difficult as a result of the density of the available data, and its frequent lack of logical and coherent organization. Quantitative hemodynamic models of the type developed here can provide a rational structure around which to present the data to clinicians, providing the basis for more sophisticated and sensitive alarms, and playing a pivotal role in developing decision support paradigms to guide therapy. Computational models of cardiovascular function are also valuable educational tools for students in high school, college, university, and medical school.
Task Progress:	During the third year of funding, our effort revolved around enhancing the cardiovascular simulator and validating its transient response during postural stress. We have also begun performing sensitivity analyses with particular interest in estimating parameters of the cardiovascular system from the transient hemodynamic response to standing-up. Below, we itemize the progress in each of these areas: Transient Response: To understand better the transient hemodynamic events elicited during transitions from the supine to the head-up posture (or vice-versa), we previously conducted a small clinical study in which ten normal healthy volunteers underwent an active change in posture (stand-up), a slow head-up tilt (HUT to 75 degrees over 55 seconds), and a fast head-up tilt (HUT to 75 degrees over 2 seconds). Arterial blood pressure and heart rate were recorded continuously. Simulation results were obtained by matching the model's response to the changes in mean arterial pressure and heart rate at the onset of rapid tilt. Once matched to these two time series, the remaining six transient responses (HR and MAP response at the conclusion of rapid tilt, HR and MAP response at the beginning of slow tilt, and HR and MAP response at conclusion of slow tilt) were then predicted using the same parameter settings. The simulations are capable of reproducing all major features of the transient response within the limits of experimental observations. Furthermore, the simulations are consistent in that matching the model's response to one stress (rapid tilt) allows us to predict the outcome of a different stress (slow tilt). Parameter Estimation: Major effort was allocated to implementing an algorithm that allows for parameter estimation by minimizing an error criterion between model output and hemodynamic time series from astronauts and bed-rest subjects. While other error criteria can be envisioned and easily tested, we chose to use a least-squares error criterion to measure the distance between the time series. Our model is quite non-linear with a large number of parameters that can be optimized in order to fit a given data set. In fact, when optimizing over all parameters it is very likely that the resultant parameter estimates are very sensitive to small perturbations of the data --- a phenomenon that is typical of ill-conditioned problems. To circumvent this problem, we implemented a subset selection algorithm for gradient-based, non-linear least squares parameter estimation. Subset selection allows for partitioning of the model parameters into well-conditioned and ill-conditioned parameters. The former are likely to be estimated reliably from the given data streams whereas parameter estimates of the latter are likely to be unreliable, and their presence makes the parameter estimation problem very sensitive. Application of this algorithm to simulated data allowed for recovery of the true parameter values to within 5% of their nominal v [truncated]
Bibliography Type:	Description: (Last Updated: 08/08/2019)
Articles in Peer-reviewed Journals	Mukkamala R, Toska K, Cohen RJ. "Noninvasive identification of the total peripheral resistance baroreflex." Am J Physiol Heart Circ Physiol. 2003 Mar;284(3):H947-59. PMID: 12433656 , Mar-2003
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